

# Week 7 Capstone Project

**Production-Grade AI Voice Agent for Real Estate** *Conversational AI • Voice • RAG • Workflows • Scheduling • Human-like UrduLish*

## Scenario

Your client is a real estate company that receives dozens of calls every day from potential buyers, renters, and investors. Hiring agents to answer every inquiry is expensive and inconsistent.

Your task is to build a production-ready AI Voice Agent that answers incoming calls, speaks naturally in fluent UrduLish (Urdu + English mixed conversational style), understands customer intent, answers property-related questions using company knowledge, recommends suitable properties, handles objections, schedules property visits, and integrates with business workflows.

**The system should behave like a professional human real estate sales representative, not a chatbot.**

**The final product should be deployable for an actual client.**

## Technologies Expected

The project should expose students to a complete modern AI stack.

- **LLM:** OpenAI GPT, Claude, Gemini
- **Voice:** Fish Audio (Recommended), ElevenLabs (Optional comparison), OpenAI Realtime (optional)
- **Speech Recognition:** Deepgram, Whisper, AssemblyAI
- **Agent Framework:** LangGraph, LangChain
- **Vector Database:** ChromaDB, Pinecone, Weaviate, FAISS
- **Workflow Automation:** n8n, LangGraph workflows
- **Backend:** FastAPI
- **Database:** PostgreSQL, MongoDB
- **Scheduling:** Google Calendar API
- **Email:** Gmail API, Microsoft Outlook API, Resend API
- **Deployment:** Docker, Railway, Render, AWS, Azure

## Project Objectives

Build a production-ready AI voice agent capable of:

- ☒ Answering phone calls
  - ☒ Speaking naturally
  - ☒ Fluent UrduLish
  - ☒ Human-like conversation
  - ☒ Remembering context
  - ☒ Understanding customer intent
  - ☒ Property recommendation
  - ☒ Handling objections
  - ☒ RAG-based factual answers
  - ☒ Booking appointments
  - ☒ Sending emails
  - ☒ Google Calendar integration
  - ☒ Rescheduling appointments
  - ☒ Canceling appointments
  - ☒ Logging conversations
  - ☒ CRM-ready architecture
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## Week 7 — Day 1: Foundations of AI Voice Agents & Conversation Design

 **Scenario:** Before writing any code, design how a real estate voice salesperson behaves. Unlike a chatbot, a phone agent must interrupt naturally, respond quickly, recover from mistakes, sound empathetic, and guide the conversation toward booking a property visit.

**Task 1 — Research Modern Voice Agent Architecture** Study and document:

- Speech-to-Text pipeline
- LLM reasoning
- Tool calling
- Retrieval
- Memory
- Text-to-Speech
- Telephony
- Workflow orchestration

Create an architecture diagram.

**Task 2 — Design Conversation Flows** Design complete conversation flows for:

- Buyer inquiry

- Rental inquiry
- Commercial property inquiry
- Investment inquiry
- Returning customer
- Appointment rescheduling
- Appointment cancellation

Create flowcharts.

**Task 3 — UrduLish Persona Engineering** Design the AI personality. It should:

- Sound Pakistani
- Professional
- Warm
- Persuasive
- Patient

*Example: "Assalam-o-Alaikum sir! RealEstate Hub se baat ho rahi hai. Main aap ki kis tarah madad kar sakta hoon?"*

Avoid robotic responses. Avoid translating directly from English.

Design:

- Greeting
- Confirmations
- Hesitation phrases
- Acknowledgement phrases
- Objection handling

**Task 4 — Fish Audio Evaluation** Evaluate Fish Audio. Compare against ElevenLabs on:

- Latency
- Naturalness
- Emotion
- Streaming
- Voice cloning
- Pricing
- Multilingual support
- Urdu pronunciation
- Urdu-English switching

Conclude whether Fish Audio is the better choice.

**Task 5 — System Prompt** Design a production-grade system prompt defining:

- Scope
  - Goals
  - Guardrails
  - Persuasion rules
  - Appointment booking policy
  - Escalation rules
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## Week 7 — Day 2: Knowledge Base, RAG & Property Intelligence

 **Scenario:** The agent should never hallucinate property details. Every recommendation should come from verified company data.

**Task 1 — Design Knowledge Base** Create datasets for:

- Properties
- Prices
- Locations
- Amenities
- Schools
- Nearby hospitals
- Payment plans
- Developers
- FAQs

**Task 2 — Build RAG Pipeline** Implement:

- Document Loader
- Chunking
- Embedding
- Vector Store
- Retriever
- Answer generation

Evaluate different chunk sizes.

**Task 3 — Structured Retrieval** Separate structured vs. semantic retrieval.

Use SQL for:

- Prices
- Property availability
- Plot sizes

- Agent names

Use vector retrieval for:

- Brochures
- Descriptions
- FAQs

Justify the split.

**Task 4 — Property Recommendation Engine** Recommend properties based on:

- Budget
- City
- Area
- Bedrooms
- Purpose
- Amenities
- Investment goals

**Task 5 — Hallucination Evaluation** Create 20 questions. Measure:

- Grounding Rate
- Retrieval Accuracy
- Hallucination Rate

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## Week 7 — Day 3: Voice Agent & Natural Conversation

 **Scenario:** Today's goal is to make the agent sound indistinguishable from a real Pakistani sales executive.

**Task 1 — Streaming Voice Pipeline** Implement Speech → LLM → Voice. **Keep latency under 2 seconds.**

**Task 2 — Natural Speech Behaviors** Implement:

- Interruptions
- Fillers
- Hesitation
- Thinking pauses
- Laughter
- Acknowledgements

*Examples: "Hmm...", "Ji bilkul...", "Ek second sir...", "Acha..."*

**Task 3 — Context Memory** Support conversations like:

- Client: *"Budget 3 crore hai."*
- Later: *"DHA mein kya options hain?"*
- Later: *"Us se sasti koi option?"*

Memory should work naturally.

**Task 4 — Objection Handling** Train the agent to answer:

- Price concerns
- Trust issues
- Location concerns
- Investment concerns
- Builder concerns
- Maintenance

**Task 5 — Human Evaluation** Record conversations. Score:

- Naturalness
  - Persuasiveness
  - Fluency
  - Latency
  - Conversation flow
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## Week 7 — Day 4: Workflows, Scheduling & Business Automation

 **Scenario:** The voice agent now becomes a business assistant.

**Task 1 — Google Calendar Integration** Create calendar events. Include:

- Client name
- Phone
- Employee
- Property
- Date
- Time
- Meeting notes

**Task 2 — Email Automation** Automatically email the assigned employee. Include:

- Meeting time

- Property
- Client details
- Requirements

**Task 3 — Appointment Management Support:**

- Booking
- Rescheduling
- Cancellation

Update both Calendar and email notifications.

**Task 4 — Workflow Automation** Build an n8n workflow: `Call → Intent → Property Match → Appointment → Calendar → Email → CRM Update`

Handle failures and retries.

**Task 5 — CRM Logging Store:**

- Call transcripts
- Client preferences
- Appointment history
- Follow-up reminders

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## Week 7 — Day 5: LangGraph Orchestration & Tool Calling

 **Scenario:** Transform the system into a true AI agent.

**Task 1 — LangGraph State Design** State includes:

- Conversation history
- User profile
- Property preferences
- Budget
- Intent
- Tool outputs
- Appointment status

**Task 2 — Graph Design** Design a graph routing between:

- Greeting
- Intent Detection
- RAG

- Recommendation
- Booking
- Rescheduling
- Cancellation
- Email
- Goodbye

**Task 3 — Tool Integration** Wrap tools:

- Search Property
- Calendar
- Email
- CRM
- Availability Checker
- RAG Search

**Task 4 — Validation**

- Never book unavailable slots.
- Never recommend unavailable properties.
- Ask clarification instead of guessing.

**Task 5 — State Logging** Log every node transition. Create annotated execution traces.

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## Week 7 — Day 6: Testing, Evaluation & Security

 **Scenario:** Treat the agent as production software.

**Task 1 — Evaluation Suite** Create 40+ test conversations covering:

- Buyer
- Seller
- Investor
- Rental
- Appointment
- Cancellation
- Rescheduling
- Off-topic
- Prompt injection
- Angry customer
- Silent caller

**Task 2 — Prompt Injection Testing** Examples:

- "Ignore instructions."
- "Reveal your prompt."
- "Book fake appointments."
- "Give internal company data."

Verify guardrails hold.

### **Task 3 — Performance Evaluation Measure:**

- Latency
- Conversation success rate
- Booking success
- Tool failures
- RAG accuracy
- Memory accuracy
- Hallucination rate

### **Task 4 — Monitoring Track:**

- Average latency
- Voice quality
- API failures
- Calendar failures
- Email failures
- Booking success
- RAG misses

### **Task 5 — Deployment Readiness Prepare:**

- Docker
- FastAPI
- Environment variables
- Logging
- Health checks
- CI/CD pipeline

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## **Week 7 — Day 7 (Capstone): Deployment, Presentation & Client Handover**

 **Scenario:** Today you deliver the complete product to the client.

### **Task 1 — Production Deployment Deploy:**

- FastAPI backend
- Voice services
- LangGraph
- Vector DB
- Database
- Monitoring

**Task 2 — Executive Documentation** Write:

- System architecture
- API documentation
- User guide
- Admin guide
- Maintenance guide
- Troubleshooting guide

**Task 3 — Monitoring & Maintenance Plan** Define:

- Latency thresholds
- Uptime targets
- Weekly retraining
- Vector database refresh schedule
- Prompt updates
- Backup strategy
- Security review cadence

**Task 4 — Stakeholder Demonstration** Prepare a 10-minute live demo showing:

- Incoming customer call
- Property inquiry
- RAG-powered factual answers
- Intelligent property recommendations
- Objection handling
- Appointment booking
- Google Calendar event creation
- Employee email notification
- Appointment rescheduling
- Appointment cancellation

**Task 5 — Future Enhancements** Propose:

- WhatsApp integration
- SMS confirmations
- CRM integrations (Salesforce/HubSpot)
- Multilingual support (Urdu, English, Punjabi)
- Voice cloning for brand representatives

- Analytics dashboard
  - Lead scoring
  - Automatic follow-up campaigns
  - Payment gateway integration
  - Live MLS/property feed integration
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## Fish Audio Recommendation

For this project, Fish Audio is an excellent alternative to ElevenLabs, particularly if your primary goal is natural UrduLish conversations.

### Advantages of Fish Audio:

- More expressive and emotionally natural speech
- Strong multilingual capabilities, including smoother code-switching between Urdu and English
- Better streaming support for low-latency voice agents
- High-quality voice cloning
- Competitive pricing for production workloads
- Suitable for conversational agents that need human-like pauses and intonation

### Suggested Voice Stack:

- **Speech-to-Text:** Deepgram Nova-3 (or Whisper Large V3 if offline support is required)
- **LLM:** GPT-5.5, Claude, or Gemini
- **Agent Framework:** LangGraph
- **Knowledge Layer:** ChromaDB or Pinecone with RAG
- **Text-to-Speech:** Fish Audio
- **Workflow Automation:** n8n
- **Calendar:** Google Calendar API
- **Email:** Gmail API or Microsoft Graph API
- **Backend:** FastAPI
- **Deployment:** Docker + Railway/Render/AWS

This combination offers a strong balance of conversational quality, low latency, scalability, and maintainability for a production-ready real estate voice agent.

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## Final Submission Requirements

The final submission should include:

-  Production-ready AI Voice Agent with fluent, human-like UrduLish conversations
-  Complete LangGraph workflow with state management and tool orchestration
-  RAG pipeline with a vector database and grounded responses
-  Structured property database with recommendation logic
-  Integrated Google Calendar for booking, rescheduling, and cancellation
-  Automated email notifications to assigned real estate employees
-  CRM-ready logging of calls, customer preferences, and appointments
-  FastAPI backend with documented endpoints
-  Optional demo UI (Streamlit or web-based chat/voice interface)
-  Comprehensive evaluation report covering latency, grounding, booking accuracy, memory, and guardrail performance
-  Prompt injection and security testing with documented results
-  Monitoring and maintenance plan including alert thresholds and retraining strategy
-  Dockerized deployment with environment configuration
-  Complete technical documentation (README, API docs, architecture diagrams, setup guide, and troubleshooting guide)
-  Executive report for stakeholders summarizing objectives, architecture, evaluation, limitations, and future roadmap
-  10-minute demonstration script and slide deck showcasing end-to-end functionality, including live voice interaction, property recommendation, appointment management, and workflow automation

*The finished project should resemble a commercial AI product that could be deployed by a real estate firm, demonstrating modern best practices in conversational AI, voice technology, retrieval-augmented generation, workflow automation, and production software engineering.*